

Data Science (CSE558)

End-Sem

1. Let A and B be two events from the same sample space such that $\Pr(A) = \frac{1}{x}$ and $\Pr(B) = \frac{x}{x+1}$, for some $x > 1$. Prove or disprove that event A and B are disjoint. (3 Marks)

Answer

Note that $\Pr(A) + \Pr(B) = \frac{1}{x} + \frac{x}{x+1} = \frac{x+1+x^2}{x(x+1)} = 1 + \frac{1}{x+1} > 1$. The sum of disjoint events are always less or equal to 1. Hence, the events A and B are not disjoint.

Rubric

Award 2 marks for showing $\Pr(A) + \Pr(B) > 1$.

Award 1 mark for correct argument why the events are not disjoint.

2. Suppose there are five events E_1, E_2, E_3, E_4, E_5 (need not be independent) with probability of their occurrences are p_1, p_2, p_3, p_4, p_5 . Using union bound give a tight lower bound to the probability that none of the five events occur. (3 Marks)

Answer

The probability that one of the event happens is $\Pr(E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5) \leq p_1 + p_2 + p_3 + p_4 + p_5$. Hence, probability that none of the event happens is $1 - \Pr(E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5) \geq 1 - (p_1 + p_2 + p_3 + p_4 + p_5)$.

3. Let a company claims that it has 10,100 images of unique captchas. To test it, you have requested m images that are sampled uniformly at random, with replacement from the corpus of 10,100 images. Now compute the largest value of m so that you see at least one captcha repeated in the set of m random samples with at most 50% probability? (5 Marks)

Answer

If there are m samples then there are $\binom{m}{2} = \frac{m(m-1)}{2}$ possible repetitions. Now consider $\binom{m}{2}$ indicator random variables (say r) which are 0 if no repetitions, else 1. The $\binom{m}{2}$ trial returns $\{r_1, r_2, r_3, \dots, r_{\binom{m}{2}}\}$ samples from r . Let $R = \sum_{i=1}^{\binom{m}{2}} r_i$. Now, $R \geq 1$ implies that there is at least one repetition and $E[R] = \binom{m}{2} E[r]$.

Note that $E[r] = \Pr(\text{two images are same}) = \frac{1}{10,100}$. Now, using Markov's inequality we have,

$$\Pr(R \geq 1) \leq \frac{\binom{m}{2} E[r]}{1} = \frac{\binom{m}{2}}{10,100} \leq \frac{1}{2}$$

Now, solving the above equation we get, $m \leq 101$.

Rubric

Award 1 mark probability (or expectation) of same image (or collision) for a fixed pair is $\frac{1}{10,100}$

Award 2 marks for correctly applying Markov's inequality.

Award 2 marks for correctly analyzing value of m

4. Answer the following questions on hypothesis testing.

- a) Recall t-test for population mean. For a given sample, if its mean is p and it is assumed that the population mean is p_0 , then state alternate hypothesis (H_1) and null hypothesis (H_0), to test, (i) two tailed test, (ii) right tailed test and (iii) left tailed test. (+2/-1 Marks)

Answer

Let μ be the population mean.

- (i) $H_0: \mu = p_0$ & $H_1: \mu \neq p_0$
- (ii) $H_0: \mu \leq p_0$ & $H_1: \mu > p_0$
- (iii) $H_0: \mu \geq p_0$ & $H_1: \mu < p_0$

Rubric:

Ignore H0 and any variable can be used in place of μ .

Full grade if all the H1 are correct, if incorrect then -1. If not attempted 0.

- b) It is believed that more households prefer OTT platforms than cable/Dish connections on their TVs. From a survey, it has been observed that 43 families prefer OTT platforms out of 100. With a 5% significance level, do you think that the above belief is incorrect? (3 Marks)

Answer

Let, μ be the proportion of households who watches OTT.

H0: $\mu > 0.5$ & H1: $\mu \leq 0.5$

$$z = \frac{0.43 - 0.5}{\sqrt{\frac{0.5 \cdot 0.5}{100}}} = -1.4$$

Since, z-statistics > -1.64. Hence, we do not reject H0.

Rubric

Award 1 mark for correct H1 statement.

Award 1 mark for correct computation of z-statistics

Award 1 mark for final conclusion.

5. Consider a hypercube of edge length 1 in a d -dimensional space such that the center of the hypercube is at the origin and a hypersphere of unit radius whose center is again at the origin. In 2-dimensional space hypercube and hypersphere are commonly known as square and circle respectively. In 3-dimensional space the hypercube and hypersphere are commonly known as cube and sphere respectively.

- a) Compute the value of d , such that in the d -dimensional space the corners of the hypercube are on the surface of the hypersphere. (3 Marks)

Answer

Note, that the surface of a unit radius hypersphere is made up of unit norm vectors. Next, the unit length hypercube, whose origin is at the center has corners at $\{-\frac{1}{2}, +\frac{1}{2}\}^d$. Hence, the corners are at a

distance $r = \sqrt{\sum_{i=1}^d \left(0 - \frac{1}{2}\right)^2} = \frac{\sqrt{d}}{2}$ from the origin. Now at $d = 4$, we have $r = 1$. Hence, the corners of a hypercube at $d = 4$ are on the surface of the hypersphere.

Rubric

Award 2 marks if the distance of corner or half of the largest diagonal are computed.

Award 1 mark if the final answer is 4 after calculations.

- b) Compute the value of d , such that in the d -dimensional space the corners of the hypercube are outside the surface of the hypersphere. (2 Marks)

Answer

At, $d \in (4, \infty)$, the corners of the hypercube lies outside the hypersphere.

Rubric

Award 2 marks if correct.

6. Let $x_1, x_2, \dots, x_n$ are iid (independent, identically distributed) samples from a random variable r whose mean is 10 and standard deviation is 5. Let $X = \frac{1}{n} \sum_{i=1}^n x_i$.

a) What is the smallest value of n such that $\Pr(X > 12) \leq 0.2$? (3 Marks)

Answer

Notice, that by applying Markov's inequality we $\Pr(X > 12) \leq 5/6$, which is undesirable. Hence, we will Chebyshev's inequality.

$$\Pr(|X - 10| > 2) \leq \frac{25}{4n}$$

Now, to ensure that $\frac{25}{4n} \leq 0.2$, we need $n \geq 32$.

Rubric

Award 2 marks for correctly applying Chebyshev's

Award 1 mark for computing $n \geq 32$.

b) Prove Law of Large Number, i.e., $\lim_{n \rightarrow \infty} X = 10$. Hint: Use Chebyshev's inequality and show that the following is true for any arbitrary $\varepsilon > 0$, (2 Marks)

$$\lim_{n \rightarrow \infty} \Pr(|X - E[r]| > \varepsilon) = 0$$

Answer

We know, that for a given random variable Y and any constant c , $\text{Var}[cY] = E[(cY)^2] - (E[cY])^2 = c^2 \cdot \text{Var}[Y]$. So, due to independence we have the following.

$$\text{Var}[X] = \sum_{i=1}^n \text{Var}\left[\frac{X_i}{n}\right] = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

So, applying Chebyshev's inequality on X for any $\varepsilon > 0$ we get.

$$\mathbf{P}(|X - \mathbf{E}[X]| \geq \varepsilon) \leq \frac{\sigma^2}{n \cdot \varepsilon^2}$$

In the LHS as $n \rightarrow \infty$, except for n , everything else (i.e., σ^2 and ε^2) are constant, so as n increases the LHS decreases.

$$\lim_{n \rightarrow \infty} \mathbf{P}(|X - \mathbf{E}[X]| \geq \varepsilon) \rightarrow 0$$

Rubric

Award 1 mark for computing $\text{Var}[X]$ that depends on n^{-1}

Award 1 mark for showing the final claim when $n \rightarrow \infty$.

7. Consider you have a biased coin. You toss it for a pair of 4 times and observed (HH, HH, HT, TH). Let θ be the probability of getting heads. Answer the following.

a) Let the probability of getting head is restricted, $\theta = \{0.3, 0.5, 0.7\}$. From the restriction, which is most likely θ_{MLE} ? (1 Marks)

Answer

As $L = p^6 \cdot (1 - p)^2$, so, at $p = 0.3$ we get, $L = 0.00036$, at $p = 0.5$ we get $L = 0.0039$ and at $p = 0.7$ we get $L = 0.011$. Since, L is maximum at $p = 0.7$, so from the given restriction the most likely $\theta_{\text{MLE}} = 0.7$.

Rubric

Award full marks if correctly computed.

b) What is θ_{MLE} , the probability of getting head (MLE)? (2 Marks)

Answer

Notice, that there are a total of 8 tosses out of which there are 6 heads and 2 tails. Let the probability of getting heads is p . Then after ignoring the constant factor 8C_6 , the likelihood of the given event is,

$$L = p^6 \cdot (1 - p)^2. \text{ So, the log likelihood is } \log(L) = 6 \log(p) + 2 \log(1 - p)$$

Now, to find the MLE.

$\frac{d \log(L)}{dp} = \frac{6}{p} - \frac{2}{1-p}$. So, at $p = \frac{3}{4}$ we get, $\frac{d \log(L)}{dp} = 0$. As the second derivative is always

negative, so $\theta_{MLE} = \frac{3}{4}$, which maximizes the likelihood of the given sample.

Rubric

Award full marks if $\theta_{MLE} = \frac{3}{4}$ is reached by computing the number of heads over number of trials, else award 1 mark.

- c) Let the probability of getting head is restricted, $\theta = \{0.3, 0.5, 0.7\}$. Let $P(\theta=0.3) = 0.4$, $P(\theta=0.5) = 0.5$ and $P(\theta=0.7) = 0.1$. From this, which is most likely θ_{MAP} ? (3 Marks)

Answer

As $L = p^6 \cdot (1-p)^2$ and $\theta_{MAP} \sim L(\theta) \cdot P(\theta)$. So, at $p = 0.3$ we get, $\theta_{MAP}(0.3) = 0.000144$, at $p = 0.5$ we get $\theta_{MAP}(0.5) = 0.00195$ and at $p = 0.7$ we get $\theta_{MAP}(0.7) = 0.0011$. Since, θ_{MAP} is maximum at $p = 0.5$, so from the given restriction the most likely $\theta_{MAP} = 0.5$.

Rubric

Award full marks if $\theta_{MAP} = 0.5$ is reached by considering likelihood and posterior, else award 1 mark.

8. Recall, JL matrix of size $n \times d$ such that n -dimensional vectors are projected on to a d -dimensional space while preserving L_2 norms of the vectors.

- a) Define a random variable with appropriate mean and variance such it gives a dense JL where 100% of indices are nonzero. (2 Marks)

Answers

Let the JL matrix be M , where, every index of M is sampled from the following distribution.

$$m = \begin{cases} -\sqrt{\frac{1}{d}} & \text{with probability } 0.5 \\ +\sqrt{\frac{1}{d}} & \text{with probability } 0.5 \end{cases}$$

Show that, $E[M^T M] = I$. The important point here, is that the off diagonal terms of $M^T M$ are such that over expectation they are 0, and diagonal terms are summation of d random variables whose variances are $\frac{1}{d}$.

Rubric:

Award 1 mark for correct random variable

Award 1 mark for arguing either $E[M^T M] = I$ or showing $E[\|M^T x\|_2^2] = \|x\|_2^2$ for any $x \in \mathbb{R}^n$.

- b) Define a random variable such that it gives a sparse JL where 60% of indices are zero. Show its correctness. (3 Marks)

Answers

Let the JL matrix be M , where, every index of M is sampled from the following distribution.

$$m = \begin{cases} -\sqrt{\frac{5}{2d}} & \text{with probability } \frac{2}{10} \\ +\sqrt{\frac{5}{2d}} & \text{with probability } \frac{2}{10} \\ 0 & \text{with probability } \frac{6}{10} \end{cases}$$

Show that, $E[M^T M] = I$. The important point here, is that the off diagonal terms of $M^T M$ are such that over expectation they are 0, and diagonal terms are 1.

Rubric:

Award 1 mark for correct random variable

Award 2 mark for arguing either $E[M^T M] = I$ or showing $E[\|M^T x\|_2^2] = \|x\|_2^2$ for any $x \in \mathbb{R}^n$.

9. Let A be a matrix of size $n \times m$ such that its rank is r , where $r < \min\{m, n\}$. Let U and V be the left and right singular vectors of A and Σ be the diagonal matrix representing non-zero singular values in order, such that singular value (say σ_1) in the first row is greater than the singular value (say σ_2) in the second row, which is greater than the singular value (say σ_3) in the third row and so on.

- a) What are the dimensions of U , V and Σ , if $A = U\Sigma V^T$? (+1 / -0.5 Marks)

Answer

$A = U\Sigma V^T$, where $U \in \mathbb{R}^{n \times r}$, Σ is an $r \times r$ diagonal matrix and $V \in \mathbb{R}^{m \times r}$.

Rubric

If correctly answered then 1, if incorrectly answered then -0.5, else 0.

- b) Let, k be an integer such that $k < r$. Now compute a matrix B using the singular values and singular vectors of A such that it is the worst rank k approximation of A (i.e., instead of top k singular values it retains the bottom k singular values). Then compute $\|A - B\|_F^2$. (5 Marks)

Answer

Let, $U = [u_1, u_2, \dots, u_r] \in \mathbb{R}^{n \times r}$, where u_i is the i^{th} column of U . Now the smallest k singular values are $\sigma_{r-k+1}, \sigma_{r-k+2}, \dots, \sigma_r$ and there corresponding singular vectors in U are $u_{r-k+1}, u_{r-k+2}, \dots, u_r$. Let $U_k = [u_{r-k+1}, u_{r-k+2}, \dots, u_r] \in \mathbb{R}^{n \times k}$, then worst rank k approximation of A , is the projection of A on the space spanned by U_k . Hence,

$$B = U_k U_k^T A$$

$$\text{Now, } \|A - B\|_F^2 = \|U\Sigma V^T - U_k U_k^T U\Sigma V^T\|_F^2 = \|(I - U_k U_k^T)U\Sigma V^T\|_F^2 = \sum_{i=1}^{r-k} \sigma_i^2.$$

Rubric:

Award 2 marks for correctly defining B .

Award 3 marks for correctly bounding the term $\|A - B\|_F^2$

10. Let, $\mathcal{L}$ be the set of vectors on the surface of a unit sphere in $\mathbb{R}^n$.

- a) If $\mathcal{H}$ is the $\frac{1}{2}$ -net of the set $\mathcal{L}$, then what is the known relation between any vector in $\mathcal{L}$ and some vector in $\mathcal{H}$? (2 Marks)

Answer

For every $u \in \mathcal{L}$, there is a $v \in \mathcal{H}$, such that $\|u - v\|_2^2 \leq \frac{1}{2}$.

Rubric

Award 1 mark if $\|u - v\|_2^2 \leq \frac{1}{2}$, without mentioning condition on u and v .

Award 1 mark if the conditions are mentioned correctly.

- b) (Bonus) Recall that any vector that does not belong to the net, can be represented by infinitely many scaled vectors from the net, i.e., $v_1 = v_2 + v_3 + v_4 + \dots$. Here $v_1 \in \{\mathcal{L} \setminus \mathcal{H}\}$, i.e., v_1 is a vector on the surface that is not in the set $\mathcal{H}$ and for all the vectors in the RHS, some multiple of them belongs to $\mathcal{H}$. Define the conditions on the vectors in the RHS and argue, with a finite size $\frac{1}{2}$ -net how is it possible to get an infinite sequence. (3 Marks)

Answer

The RHS vectors are such that for any $i \geq 2$, $\|v_i\|_2^2 \leq \frac{1}{2^{i-2}}$ such that, $u_i \in \mathcal{H}$ where $u_i = 2^{i-2} v_i$.

There are infinitely many vectors because not all the vectors in the RHS need to have unique representative in $\mathcal{H}$. By definition, even a single $v \in \mathcal{H}$, can have infinitely many vectors as defined in the RHS of the above equation.

Rubric

Award 1 mark for giving a bound on the norm of v_i . Award marks if the bounds are any decreasing GP series.

Award 1 mark for showing that some scale of every RHS belongs to $\mathcal{H}$.

Award 1 mark for arguing that there are infinitely many possible vectors for RHS.

11. Let A be matrix of size $n \times d$ whose rank is r , such that $r < d < n$ and $Q := \mathbb{R}^d$. Let, $f: A \times Q \rightarrow \mathbb{R}_{\geq 0}$ such that, for every $q \in Q$ we have the following.

$$f(A, q) = \sum_{i=1}^n (A_i^T q)^2 = \|Aq\|_2^2$$

Here, A_i^T represents the i^{th} row (or datapoint) of the matrix A . Now compute a tight upper bound of the sensitivity score of i^{th} row (or datapoint) and compute the total sum of all the sensitivity scores. (5 Marks)

Answer

Sensitivity score of A_i^T is $\sup_{q \in \mathbb{R}^d} \frac{(A_i^T q)^2}{\|Aq\|_2^2}$. Now consider a fixed q .

$$\begin{aligned} \frac{(A_i^T q)^2}{\|Aq\|_2^2} &= (A_i^T q)(q^T A^T A q)^{-1}(q^T A_i) = (A_i^T q)(Aq)^+(q^T A^T)^+(q^T A_i) = A_i^T q q^+ (A^T A)^{-1} (q^T)^+ q^T A_i \\ &\leq A_i^T (A^T A)^{-1} A_i \end{aligned}$$

Since the $A_i^T (A^T A)^{-1} A_i$ is independent of q , hence, $\sup_{q \in \mathbb{R}^d} \frac{(A_i^T q)^2}{\|Aq\|_2^2} \leq A_i^T (A^T A)^{-1} A_i$.

Let $A = U \Sigma V^T$, where U_i is the i^{th} row of U .

$$\frac{(A_i^T q)^2}{\|Aq\|_2^2} \leq A_i^T (A^T A)^{-1} A_i = \|U_i\|_2^2$$

Now the sum of sensitivity scores is $\sum_{i=1}^n \|U_i\|_2^2 = \|U\|_F^2 = r$. The last equality is because rank of A is r .

Rubric:

Award 3 marks for analyzing $\sup_{q \in \mathbb{R}^d} \frac{(A_i^T q)^2}{\|Aq\|_2^2} \leq A_i^T (A^T A)^{-1} A_i$

Award 2 marks for proving/giving correct argument that total sensitivity score is r .

12. Consider weighted dataset (A, ω) and a loss function $f: A \times \omega \times X \rightarrow \mathbb{R}_{>0}$, which is minimize over the set X . Let a weighted subsample (A, v) be a strong (ε, δ) -approximation, where $\varepsilon \in (0, 1/5)$ and a small δ close to 0. Hence, we have the following inequality for two very special elements $\bar{x} \in X$ and $\bar{\bar{x}} \in X$ and a constant k with a high probability.

$$f(A, \omega, \bar{\bar{x}}) \leq (1 + k\varepsilon)f(A, \omega, \bar{x})$$

- a) State the required condition on (A, v) for it to be a strong (ε, δ) -approximation. (3 Marks)

Answer

For all $x \in X$ with at least $1 - \delta$ probability the following is true.

$$(1 - \varepsilon)f(A, \omega, x) \leq f(A, v, x) \leq (1 + \varepsilon)f(A, \omega, x)$$

Rubric:

Award 1 mark if the above inequality is not written.

Plus 1 mark if $x \in X$ or something that relatable is mentioned.

Plus 1 mark if 'at least $1 - \delta$ probability' is claimed.

- b) What are $\bar{x}$ and $\bar{\bar{x}}$ in the above setup. (2 Marks)

Answers: $\bar{x}$ and $\bar{\bar{x}}$ are the elements in X that optimizes the functions $f(A, \omega, x)$ and $f(A, v, x)$ respectively. i.e.,

$$\bar{x} \in \operatorname{argmin}_{x \in X} f(A, \omega, x) \text{ and } \bar{\bar{x}} \in \operatorname{argmin}_{x \in X} f(A, v, x).$$

Rubric: Award 1 mark for each correct definition.

- c) Show that the above inequality is true and compute an appropriate k for a constant k . (5 Marks)

Answer

$$(1 - \varepsilon)f(A, \omega, \bar{x}) \leq f(A, v, \bar{x})$$

\\ True due to strong (ε, δ) -approximation

$$f(A, v, \bar{x}) \leq f(A, v, \bar{x})$$

\\ True, because $\bar{x} \in \operatorname{argmin}_{x \in X} f(A, v, x)$.

$$f(A, v, \bar{x}) \leq (1 + \varepsilon)f(A, \omega, \bar{x})$$

\\ By strong (ε, δ) -approximation

So, we have

$$(1 - \varepsilon)f(A, \omega, \bar{x}) \leq (1 + \varepsilon)f(A, \omega, \bar{x})$$

$$f(A, \omega, \bar{x}) \leq \frac{(1 + \varepsilon)}{(1 - \varepsilon)} f(A, \omega, \bar{x})$$

Now, setting ε to be the highest possible value in the denominator we get the following

$$f(A, \omega, \bar{x}) \leq \left(1 + \frac{5\varepsilon}{2}\right) f(A, \omega, \bar{x})$$

Rubric

Award 2 marks for correct use of strong (ε, δ) -approximation

Award 1 mark for correctly mentioning $f(A, v, \bar{x}) \leq f(A, v, \bar{x})$ or something similar.

Award 2 marks for correct k value, i.e., $\frac{5}{2}$.

- d) (Bonus) Instead of minimize let the objective is to maximize the loss function $f()$ over the same the set X . Now, for the same value of ε , derive a similar relation between $f(A, \omega, \bar{x})$ and $f(A, \omega, \bar{x})$ and answer parts b) and c). Hint: Be careful with sign and approximation factor. (7 Marks)

Answer

$$\bar{x} \in \operatorname{argmax}_{x \in X} f(A, \omega, x) \text{ and } \bar{x} \in \operatorname{argmax}_{x \in X} f(A, v, x)$$

$$f(A, v, \bar{x}) \leq (1 + \varepsilon)f(A, \omega, \bar{x})$$

\\ True due to strong (ε, δ) -approximation

$$f(A, v, \bar{x}) \leq f(A, v, \bar{x})$$

\\ True, because $\bar{x} \in \operatorname{argmax}_{x \in X} f(A, v, x)$.

$$(1 - \varepsilon)f(A, \omega, \bar{x}) \leq f(A, v, \bar{x})$$

\\ By strong (ε, δ) -approximation

So, we have

$$(1 - \varepsilon)f(A, \omega, \bar{x}) \leq (1 + \varepsilon)f(A, \omega, \bar{x})$$

$$\frac{(1 - \varepsilon)}{(1 + \varepsilon)} f(A, \omega, \bar{x}) \leq f(A, \omega, \bar{x})$$

Now, setting ε to be the highest possible value in the denominator we get the following

$$\left(1 - \frac{5\varepsilon}{3}\right) f(A, \omega, \bar{x}) \leq f(A, \omega, \bar{x})$$

Rubric: Similar to b) and c)

Formula sheet

z-tests (known variance & proportional)

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

TABLES

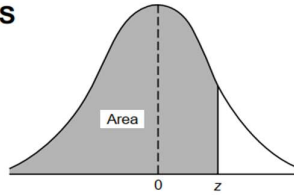


Table 1 The normal curve

(a) Area under the normal curve

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0124	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

$$Z_{0.05} = -1.645$$

$$Z_{0.0018} = -2.91$$

- $-2.91 = -(2.9 + 0.01)$, i.e, In the above table, the first column with value -2.9 and first row with 0.01.
- So, at 0.18% level of significance, the left curve critical value is -2.91.
- Since Z-scores follows a normal distribution, so you may use its property of symmetry.

Conditional Probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Law of Total

Probability: $P(B) = \sum_{i=1}^n P(B \cap A_i) = \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$

Let $X_1, X_2, \dots, X_n$ are random samples that corresponds to your n independent experiments.

Bernoulli's distribution $P(X_i|\theta) = \theta^{X_i} \cdot (1 - \theta)^{1-X_i}$

Binomial distribution $P(X_i|\theta) = \binom{n}{X_i} \theta^{X_i} \cdot (1 - \theta)^{n-X_i}$

Geometric distribution $P(X_i|\theta) = (1 - \theta)^{X_i-1} \theta$

Poisson distribution $P(X_i|\theta) = \frac{\theta^{X_i} e^{-\theta}}{X_i!}$

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χ^2 -test

$$\chi^2 = \sum_{i=1}^K \frac{(O_i - E_i)^2}{E_i}$$

v: Degree of freedom

a: Left critical value.

b: Right critical value.

	Level of significance α							
	0.20		0.10		0.05		0.01	
Two-sided	0.10		0.05		0.025		0.005	
One-sided								
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10 ⁻⁴	3.84	98.10 ⁻⁵	5.02	16.10 ⁻⁵	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

t-test

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n - 1}$$

Degrees of freedom	Significance level						
	Two-tailed test: One-tailed test:	10% 5%	5% 2.5%	2% 1%	1% 0.5%	0.2% 0.1%	0.1% 0.05%
1		6.314	12.706	31.821	63.657	318.309	636.619
2		2.920	4.303	6.965	9.925	22.327	31.599
3		2.353	3.182	4.541	5.841	10.215	12.924
4		2.132	2.776	3.747	4.604	7.173	8.610
5		2.015	2.571	3.365	4.032	5.893	6.869
6		1.943	2.447	3.143	3.707	5.208	5.959
7		1.894	2.365	2.998	3.499	4.785	5.408
8		1.860	2.306	2.896	3.355	4.501	5.041
9		1.833	2.262	2.821	3.250	4.297	4.781
10		1.812	2.228	2.764	3.169	4.144	4.587
11		1.796	2.201	2.718	3.106	4.025	4.437
12		1.782	2.179	2.681	3.055	3.930	4.318
13		1.771	2.160	2.650	3.012	3.852	4.221
14		1.761	2.145	2.624	2.977	3.787	4.140
15		1.753	2.131	2.602	2.947	3.733	4.073
16		1.746	2.120	2.583	2.921	3.686	4.015
17		1.740	2.110	2.567	2.898	3.646	3.965
18		1.734	2.101	2.552	2.878	3.610	3.922
19		1.729	2.093	2.539	2.861	3.579	3.883
20		1.725	2.086	2.528	2.845	3.552	3.850
21		1.721	2.080	2.518	2.831	3.527	3.819
22		1.717	2.074	2.508	2.819	3.505	3.792
23		1.714	2.069	2.500	2.807	3.485	3.768
24		1.711	2.064	2.492	2.797	3.467	3.745
25		1.708	2.060	2.485	2.787	3.450	3.725
26		1.706	2.056	2.479	2.779	3.435	3.707
27		1.703	2.052	2.473	2.771	3.421	3.690
28		1.701	2.048	2.467	2.763	3.408	3.674
29		1.699	2.045	2.462	2.756	3.396	3.659
30		1.697	2.042	2.457	2.750	3.385	3.646
32		1.694	2.037	2.449	2.738	3.365	3.622
34		1.691	2.032	2.441	2.728	3.348	3.601
36		1.688	2.028	2.434	2.719	3.333	3.582
38		1.686	2.024	2.429	2.712	3.319	3.566
40		1.684	2.021	2.423	2.704	3.307	3.551
42		1.682	2.018	2.418	2.698	3.296	3.538
44		1.680	2.015	2.414	2.692	3.286	3.526
46		1.679	2.013	2.410	2.687	3.277	3.515
48		1.677	2.011	2.407	2.682	3.269	3.505
50		1.676	2.009	2.403	2.678	3.261	3.496
60		1.671	2.000	2.390	2.660	3.232	3.460
70		1.667	1.994	2.381	2.648	3.211	3.435
80		1.664	1.990	2.374	2.639	3.195	3.416
90		1.662	1.987	2.368	2.632	3.183	3.402
100		1.660	1.984	2.364	2.626	3.174	3.390

Maximum Likelihood Estimation (MLE): $\theta_{MLE} \in \arg \max_{\theta} \prod_{i=1}^n P(X_i | \theta)$

Maximum a Posteriori (MAP): $\theta_{MAP} \in \arg \max_{\theta} P(\theta | X_1, \dots, X_n) = \arg \max_{\theta} \prod_{i=1}^n P(X_i | \theta) \cdot P(\theta)$

Markov's Inequality: Let X be a random variable that takes non negative values. Then for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's Inequality: Let X be a random variable. Then for any $a > 0$,

$$P(|X - E[X]| \geq a) \leq \frac{\text{Var}[X]}{a^2}$$

Hoeffding's Inequality: Let $X_1, X_2, \dots, X_n$, be n independent random variable such that $a_i \leq X_i \leq b_i$. Let $S = X_1 + X_2 + \dots + X_n$. Then for all $t > 0$,

$$P(|S - E[S]| \geq t) \leq 2 \cdot \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Sensitivity Score: Let D represents n datapoints and X be a set of elements (aka models). Let $f: D \times X \rightarrow \mathbb{R}_{\geq 0}$ be some non-negative function such that, for any $x \in X$, $f(D, x) = \sum_{i=1}^n f(D_i, x)$, where D_i is the i^{th} datapoint in D . Then, for every $i \in [n]$, its importance is measured by its sensitivity score defined as follows.

$$\sup_{x \in X} \frac{f(D_i, x)}{f(D, x)}$$